

## **When Harry Markowitz Meets Ben Graham – A Project of Optimal Equity Allocation**

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### **1. Introduction**

Mr. Li Lu was born in 1966 in Tangshan, China and settled in the U.S. to attend Columbia University, earning a B.A. in Economics, a J.D., and an M.B.A. simultaneously. Inspired by a lecture from Warren Buffett, Li began his investment career and founded Himalaya Capital in 1997 based out of Seattle, WA. Currently, the fund has an AUM of US\$14,000,000,000+ and its goal is to achieve long-term capital appreciation while minimizing the risk of permanent capital impairment. Himalaya Capital accomplishes this by identifying and acquiring global securities with substantial "economic moats" and long-term growth potential primarily in China, The United States, Japan, and South Korea, usually at a deep discount to their intrinsic value.

### **2. Portfolio Overview**

From their last 13F filing, Himalaya Capital holds a portfolio of eight stocks. Their largest individual position is Bank Of America Corporation, which represents about 29% of their portfolio. However, Himalaya Capital's largest exposure is from holding Alphabet Inc. as they hold both class A and C shares in the company, totaling a position size of around 39% in the company. Below are the rest of their portfolio positions, weightings and values.

| Stock                          | Ticker | Industry / Sector | Shares Held       | Market Value           | % of Portfolio | Rank | % Ownership in Company | Est. Avg Price (Cost Basis) |
|--------------------------------|--------|-------------------|-------------------|------------------------|----------------|------|------------------------|-----------------------------|
| Bank Of America Corp.          | BAC    | FINANCE           | 18,081,133        | \$794,665,795          | 29.31%         | 1    | 0.24%                  | \$33.11                     |
| Alphabet Inc. Class C          | GOOG   | COMMUNICATIONS    | 3,044,000         | \$579,699,360          | 21.38%         | 2    | 0.05%                  | \$90.40                     |
| Alphabet Inc. Class A          | GOOGL  | COMMUNICATIONS    | 2,543,300         | \$481,446,690          | 17.76%         | 3    | 0.04%                  | \$106.54                    |
| Berkshire Hathaway Inc Class B | BRK.B  | FINANCE           | 897,749           | \$406,931,667          | 15.01%         | 4    | 0.07%                  | \$281.34                    |
| East West Bancorp, Inc         | EWBC   | FINANCE           | 2,776,351         | \$265,863,372          | 9.81%          | 5    | 2.00%                  | \$65.59                     |
| Apple Inc                      | AAPL   | IT                | 317,700           | \$79,558,434           | 2.93%          | 6    | 0.00%                  | \$119.01                    |
| Occidental Petroleum Corp      | OXY    | ENERGY            | 1,466,500         | \$72,459,765           | 2.67%          | 7    | 0.16%                  | \$63.75                     |
| Sable Offshore Corp            | SOC    | FINANCE           | 1,343,000         | \$30,754,700           | 1.13%          | 8    | 1.51%                  | \$17.26                     |
| <b>Total</b>                   |        |                   | <b>30,469,733</b> | <b>\$2,711,379,783</b> |                |      |                        |                             |

### **3. Historical Data Analysis and Calculations**

Below are the performance metrics of the individual stocks that are held by Himalaya Capital. These metrics were calculated from a sample of the last 1,500 daily closing prices, an assumed risk-free rate of 3%, 252 trading days per year and the S&P500 as the benchmark. The daily returns for each stock can be found on the excel sheet attached to the submission. The best performing stocks in the portfolio on an annual basis are AAPL, GOOG and GOOGL.

| Metrics:                           | BAC   | GOOG  | GOOGL | BRK.B | EWBC  | AAPL  | OXY   | SOC   |
|------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|
| Avg. Daily Return (%)              | 0.05% | 0.09% | 0.09% | 0.07% | 0.07% | 0.12% | 0.04% | 0.07% |
| Avg. Daily Standard Deviation (%)  | 2.2%  | 2.0%  | 2.0%  | 1.3%  | 2.7%  | 2.0%  | 3.7%  | 2.2%  |
| Avg. Annual Return (%)             | 13.1% | 24.7% | 24.3% | 18.7% | 20.0% | 35.9% | 11.6% | 19.2% |
| Avg. Annual Standard Deviation (%) | 34.2% | 31.5% | 31.6% | 21.0% | 43.2% | 31.9% | 58.3% | 35.1% |
| Beta (versus S&P500 Benchmark)     | 1.18  | 1.13  | 1.13  | 0.79  | 1.30  | 1.22  | 1.31  | 0.22  |
| Sharpe Ratio                       | 0.29  | 0.69  | 0.67  | 0.75  | 0.39  | 1.03  | 0.15  | 0.46  |
| Information Ratio                  | -0.09 | 0.47  | 0.45  | 0.18  | 0.24  | 1.01  | -0.17 | 0.20  |
| Treynor Ratio                      | 0.09  | 0.19  | 0.19  | 0.20  | 0.13  | 0.27  | 0.07  | 0.74  |

When comparing Himalaya Capital's portfolio to the S&P 500, we see that the portfolio was overweight in the Financials and Communication Services sector by approximately 41% and 30% and underweight in the Information Technology and Energy sector by approximately 27% and 1%.

|                        | Portfolio | SPY    | Relative Weight |
|------------------------|-----------|--------|-----------------|
| Finance                | 55.26%    | 14.21% | 41.05%          |
| Communication Services | 39.14%    | 9.33%  | 29.81%          |
| Information Technology | 2.93%     | 31.00% | -28.07%         |
| Energy                 | 2.67%     | 3.66%  | -0.99%          |
| Healthcare             | 0.00%     | 11.19% | -11.19%         |
| Consumer Discretionary | 0.00%     | 10.35% | -10.35%         |
| Consumer Staples       | 0.00%     | 6.03%  | -6.03%          |
| Industrials            | 0.00%     | 7.46%  | -7.46%          |
| Utilities              | 0.00%     | 2.72%  | -2.72%          |
| Real Estate            | 0.00%     | 2.27%  | -2.27%          |
| Materials              | 0.00%     | 1.79%  | -1.79%          |

**4. Below is the correlation matrix between each of the stocks in the portfolio and the benchmark:**

| Correlation Matrix |      |      |       |       |      |      |      |      |               |
|--------------------|------|------|-------|-------|------|------|------|------|---------------|
|                    | BAC  | GOOG | GOOGL | BRK.B | EWBC | AAPL | OXY  | SOC  | SPY Benchmark |
| BAC                | 1.00 |      |       |       |      |      |      |      |               |
| GOOG               | 0.41 | 1.00 |       |       |      |      |      |      |               |
| GOOGL              | 0.41 |      | 1.00  |       |      |      |      |      |               |
| BRK.B              | 0.74 | 0.49 | 0.48  | 1.00  |      |      |      |      |               |
| EWBC               | 0.79 | 0.37 | 0.37  | 0.62  | 1.00 |      |      |      |               |
| AAPL               | 0.43 | 0.64 | 0.64  | 0.55  | 0.38 | 1.00 |      |      |               |
| OXY                | 0.54 | 0.27 | 0.27  | 0.47  | 0.47 | 0.28 | 1.00 |      |               |
| SOC                | 0.11 | 0.08 | 0.09  | 0.10  | 0.11 | 0.10 | 0.08 | 1.00 |               |
| SPY Benchmark      | 0.71 | 0.74 | 0.74  | 0.78  | 0.62 | 0.79 | 0.46 | 0.13 | 1.00          |

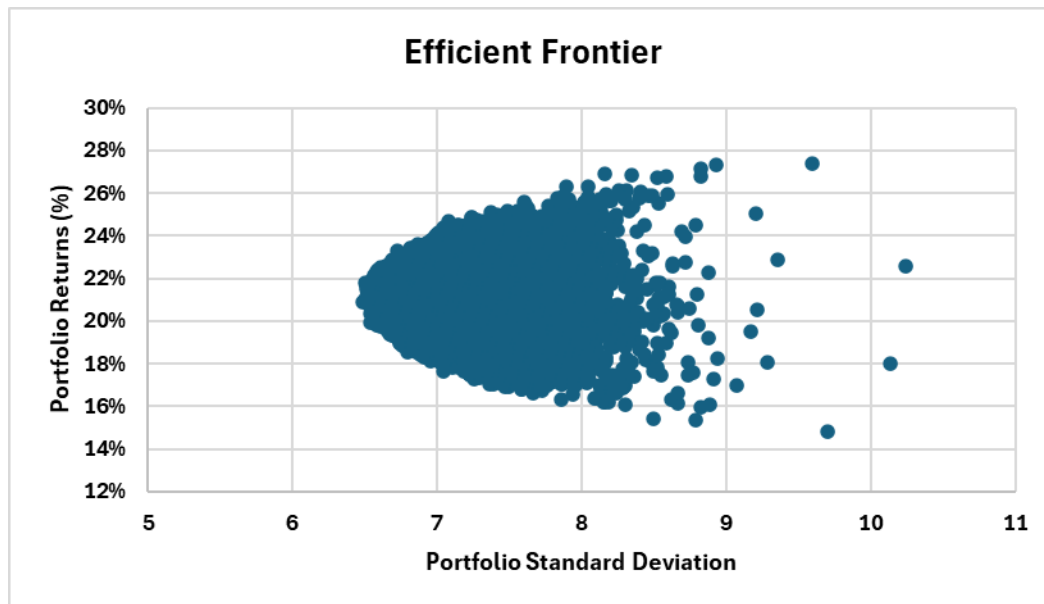
**5. Markowitz Optimization and Comparison Table**

When analyzing the portfolio and reconstructing it using the Markowitz optimization theorem, we utilized Excel's data analytics pack and discovered notable differences between Himalaya's weighting and Markowitz weightings. Below is a table showing Himalaya's actual portfolio weightings and the Markowitz Optimal Portfolio Weights, along with the annualized average returns and the corresponding Sharpe ratios for each portfolio.

|              | Actual    | Markowitz         |            |
|--------------|-----------|-------------------|------------|
| Holdings     | Portfolio | Optimal Portfolio | Difference |
| BAC          | 29.31%    | 4.30%             | -25.01%    |
| GOOG         | 21.38%    | 15.38%            | -6.00%     |
| GOOGL        | 17.76%    | 14.94%            | -2.82%     |
| BRK.B        | 15.01%    | 11.38%            | -3.63%     |
| EWBC         | 9.81%     | 10.98%            | 1.17%      |
| AAPL         | 2.93%     | 26.75%            | 23.82%     |
| OXY          | 2.67%     | 2.36%             | -0.31%     |
| SOC          | 1.13%     | 13.92%            | 12.78%     |
| Returns      | 19.77%    | 24.86%            |            |
| Sharpe Ratio | 0.57      | 0.95              |            |

As you can see the three largest differences between the weightings of Himalaya's portfolio and the Markowitz Optimal portfolio are BAC, AAPL, and SOC. Our analysis of the differences between the two portfolios continues in [section six](#) of this paper.

Below is our graphical interpretation of the efficient frontier. We ran 10,000 simulations where we randomized the portfolio weightings. This produced 10,000 unique expected return and standard deviation values which we used to construct and graph the efficient frontier of Himalaya's current portfolio.



## 6. Comparison Discussion and Fundamental Analysis of the Securities

To better understand the difference in weightings between the current portfolio and the Markowitz optimized portfolio, we conducted a fundamental analysis on each of the securities and provided our own insights as to why the portfolio manager may have deviated from following the Markowitz optimal portfolio.

We believe that Himalaya's current portfolio reflects a concentrated, fundamentals-driven strategy focused on higher-quality businesses with strong balance sheets, durable cash flows, and attractive valuations. The allocations also suggest a disciplined long-term approach, blending higher positions in relatively safer core compounders (such as Bank of America, Alphabet, Berkshire) with selective smaller value, speculative, and special situation plays (such as Occidental Petroleum and Sable Offshore). In addition, the portfolio appears to be heavily weighted toward financials and technology, aligning with Himalaya's strategic tilt toward sectors with strong cash generation, capital efficiency, and long-term structural advantages. With that said, the more speculative and smaller bets appear to be in the energy sector.

Below is a more in-depth analysis of each holding within Himalaya's current portfolio:

### *Bank of America (BAC)*

- Current portfolio weighting: 29.31%
- Markowitz weighting: 4.30%

- Difference: -25.01%
- BOA has a robust liquidity position (\$985bn including cash and short-term investments) versus total debt (\$757bn)
- Currently trades at ~9.5x NTM P/E, significantly below its 1-year average of ~11.5x. The valuation chart shows recent compression in valuation, possibly due to macro headwinds, presenting a value opportunity.
- LTM ROE of 9.5% and 28.6% net margin show decent profitability for a large bank.
- **Himalaya PM's Likely View:** Li Lu likely sees BAC as a high-quality, undervalued compounder with a safe balance sheet and solid earnings power, warranting a large overweight.
- **Reason for the differentiation:** The model views BAC as a high-volatility stock with lower risk-adjusted return. The PM's concentration introduces unsystematic risk into the portfolio, which is not favored by the Markowitz model, therefore in the optimized portfolio, the position size is reduced significantly.

#### *Alphabet (GOOG + GOOGL)*

- Current portfolio weighting (combined): ~39.14%
- Markowitz weighting: ~30.87%
- Difference: -8.27%
- Liquidity balance of ~\$95.7bn, 28.6% margin, and 32.9% ROE indicate a very strong fundamental profile.
- 13.9% YoY revenue growth and ~\$56.6bn in LTM levered FCF show strong scalability and reinvestment capacity.
- Trades at ~17.5x NTM P/E, down from ~23x earlier in the year (as per chart). The dislocation likely reflects macro fears or AI monetization uncertainty, which may be creating a long-term entry point.
- **Himalaya PM's Likely View:** Li Lu appears to be taking advantage of the multiple contraction on a secular winner with a strong & sustainable business model.
- **Reason for the differentiation:** We believe that the model slightly underweights compared to Himalaya's PM, due to diversification benefits or marginally diminishing returns from being over-concentrated in a single firm.

#### *Berkshire Hathaway (BRK.B)*

- Current portfolio weighting: 15.01%
- Markowitz weighting: 11.38%
- Difference: -3.63%
- ~24% margin, 14.7% ROE, and \$334.2bn in cash reinforce Berkshire's role as a capital allocator's dream, especially in a challenging macro environment, when dry powder can be put to work to buy assets at a discounted price.
- P/E has steadily risen over the year, signaling growing market confidence in its diversified structure and defensive appeal.

- **Himalaya PM's Likely View:** This holding appears to be a core stabilizer in the portfolio with low downside, solid return potential, thereby mirroring Himalaya's long-term mindset.
- **Reason for the differentiation:** There is not much of a difference between the PM and the model's weightings, we believe that it is likely due to BRK.B's diversified holdings and lower perceived risk. Therefore, the model underweights the PM's position sizing to optimize exposure across other opportunities.

#### *East West Bancorp (EWBC) – 9.8% weighting*

- Current portfolio weighting: 9.81%
- Markowitz weighting: 10.98%
- Difference: +1.17%
- 48.6% net income margin and 15.7% ROE make this name one of the most efficient regional banks.
- Trades at ~9.4x P/E, slightly below the historical average. Solid earnings profile despite a volatile banking environment.
- **Himalaya PM's Likely View:** Li Lu likely views EWBC as a high-quality regional bank with strong profitability, trading at a compelling valuation. Its efficient operations and solid earnings profile justify the overweight position in a value-focused portfolio.
- **Reason for the differentiation:** The minor overweight by the model may indicate that EWBC has good risk-to-return characteristics within its sector. Hence, the model sees it as a better risk-adjusted contributor than the manager assumes.

#### *Apple Inc. (AAPL)*

- Current portfolio weighting: 2.93%
- Markowitz weighting: 26.75%
- Difference: +23.82%
- LTM ROE at 136.5%, with consistent cash generation (\$93.8bn levered FCF)
- Trading at a high ~28x P/E – this likely explains the small allocation despite its strength.
- **Himalaya PM's Likely View:** Possibly a high-conviction but price-sensitive hold; small size likely reflects valuation discipline.
- **Reason for the differentiation:** We believe that the main reason for the differentiation between the two weightings is because of concerns about its high valuation and current events. However, the optimization model rewards Apple's strong return profile and relatively stable volatility, increasing the position size to maximize risk-adjusted returns.

#### *Occidental Petroleum (OXY)*

- Current portfolio weighting: 2.67%
- Markowitz weighting: 2.36%
- Difference: -0.31%
- Net debt ~\$25bn, negative revenue growth (-5.4%), and modest 8.9% ROE suggest relatively higher idiosyncratic risk in the portfolio.
- **Himalaya PM's Likely View:** Likely a special situation/value rotation play, possibly due to energy market cyclicalities or Buffett's involvement (indirect influence).

- **Reason for the differentiation:** We believe that the reason for the negligible differentiation of both model's position size is that OXY is likely offers sector diversification and energy exposure with acceptable volatility.

#### *Sable Offshore (SOC)*

- Current portfolio weighting: 1.13%
- Markowitz weighting: 13.92%
- Difference: +12.79%
- High YoY growth (23.6%) but negative levered FCF profile and deeply negative ROE (-174%) suggest a turnaround or optionality-driven play.
- **Himalaya PM's Likely View:** Likely a small-cap asymmetric bet with low weight, high upside optionality if offshore energy trends rebound.
- **Reason for the differentiation:** We believe that the large differentiation could stem from concerns surrounding geopolitical or regulatory risk. However, we believe that the model sees strong potential for returns relative to its risk, possibly due to its low correlation to other assets. Therefore, to diversify the portfolio and enhance returns the model increased the position size.

To summarize, we believe that Himalaya's portfolio reflects the PM's conviction-based tilts, sector-specific views, and qualitative factors like macro themes, regulatory risks, or firm-specific sentiment. While Markowitz's portfolio is constructed to effectively have the greatest risk-to-return efficiency, it is purely based on quantitative data with the goal of maximizing Sharpe ratio under constraints of minimizing risk and diversification. Our key takeaway in this section is that, while Markowitz's portfolio is statistically more efficient (Sharpe Ratio = 1.55 vs. 0.57), the PM sacrificed an optimal portfolio weighting for weightings that better reflect their thematic conviction as well as downside protection that may not be captured solely in return volatility alone.

## 7. Conclusion

Reflecting on the analysis of both portfolios, if we were the portfolio manager, we believe that we would select the Markowitz optimized portfolio weightings as the foundation of our portfolio while incorporating factors based on current market dynamics and qualitative insights to finalize our portfolio.

We believe that by using the Markowitz optimized portfolio weightings as a foundation allows us to have superior risk-adjusted performance as the Markowitz portfolio exhibits a Sharpe ratio almost double that of the current portfolio. This suggests a significantly more efficient risk-return trade-off which is backed by quantitative rigor. We also like how the Markowitz approach allocates the weight purely based on data analysis. We see that the model reallocates based on higher returns per unit of risk, hence why AAPL and SOC saw a large increase in each position. This is consistent with reports from the Value Investor Club as it praises AAPL's resilient free cash flows and low beta, and SOC's European exposure acting as a low-correlated hedge.

However, following the optimizer blindly could introduce practical risks. Real-world constraints—like sector caps, liquidity requirements, and institutional mandates—aren't captured

by pure optimization. Additionally, the Markowitz model relies heavily on historical return and covariance data to construct the optimal portfolio, assuming that past performance and relationships between assets will persist. In reality, market conditions evolve, and asset behaviors can shift dramatically in response to macroeconomic changes, technological disruption, or geopolitical events, meaning a portfolio built purely on backward-looking data may be poorly positioned for the future. Moreover, forward-looking risks such as valuation compression or regional macro weaknesses (recently flagged for AAPL and GOOGL) suggest that purely historical models may misguide future positioning. Qualitative signals, like BRK.B's distressed asset acquisitions flagged in Pitchbook, also highlight opportunities that optimizers would miss without event-driven insights.

Therefore, we suggest a hybrid approach: starting with Markowitz weights as a foundation to maximize Sharpe Ratio, but introduce caps (e.g., 20% per name) and thematic overlays based on valuation gaps (e.g., EWBC still cheap vs. peers on P/B), macro hedges (e.g., OXY for energy hedge in case of oil spikes), and catalyst-driven upside (e.g., BRK.B deal-making per 13F filings). This blended model would maintain the rigor and discipline of quantitative optimization while embedding critical qualitative judgment, ultimately aiming to construct a portfolio that is both mathematically efficient and strategically resilient.

To conclude, our report highlights how a Markowitz approach to portfolio management can help portfolio managers maximize Sharpe ratios and construct efficient portfolios that have the potential to outperform. However, real-world portfolio managers must also consider qualitative factors and unforeseen events that may not be reflected in historical data. We observe that managers like Mr. Li Lu incorporate both quantitative and qualitative convictions into their stock selection and portfolio construction, with a focus not just on outperformance but on long-term capital appreciation. Most importantly, we experienced firsthand how portfolio management is influenced by a wide range of factors, both quantitative and qualitative. While one approach may outperform in certain periods, a blend of strategies is necessary for long-term success in maximizing returns and minimizing risk. In the end, we learned that portfolio management demands not only the precision of science but the vision, intuition, and resilience of true artistry.

## **8. Appendix**